

Medical Report Intelligence & Patient-Friendly Explanation System

Solution-MediBuddy

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Introduction & Problem Statement

The Situation

- Medical reports contain complex clinical terminology
- Difficult for non-medical users to interpret
- Users cannot easily identify normal vs abnormal values

The Problem

- Manual interpretation requires medical expertise
- Lack of clear explanations leads to confusion and misinterpretation
- No simple system to convert reports into understandable insights

The Opportunity

- Use AI to simplify medical report understanding
- Provide structured, easy-to-read explanations
- Help users take informed health decisions quickly

Objective & Proposed System

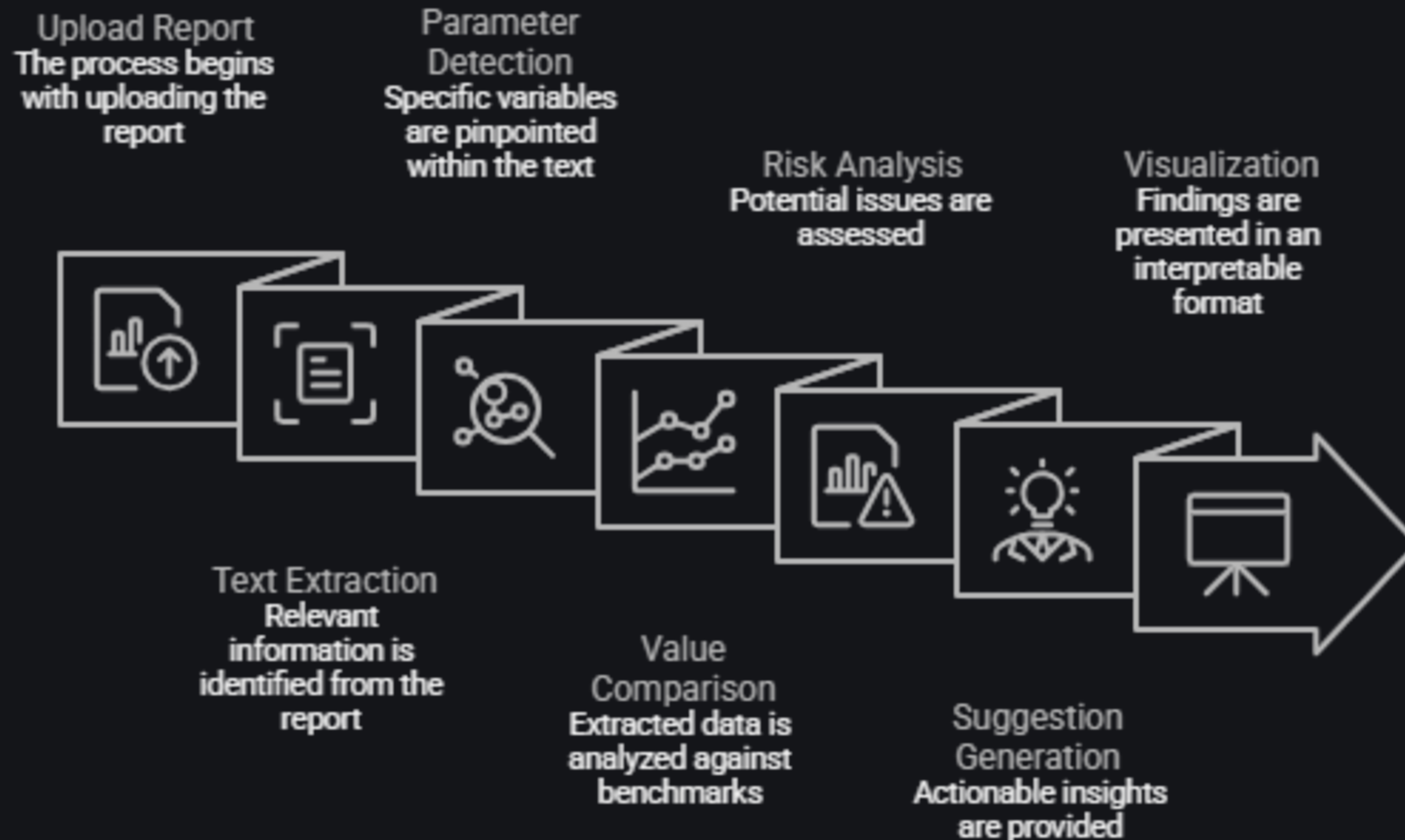
Objective

- Automatically analyze uploaded medical reports
- Extract medical parameters and values
- Compare with standard normal ranges
- Provide simple explanations and suggestions
- Enable comparison of previous and current reports

Proposed System

- Flask-based backend for processing and analysis
- React dashboard for visualization
- AI-assisted explanation using transformer models
- OCR support for scanned reports
- Structured output with risk level, alerts, and suggestions

Workflow and Tech Stack



AXIOS



FLAN-T5



Real-Time Industry Usage & Future Scope

Current Applications

- Digital health platforms
- Telemedicine applications
- Hospital patient dashboards
- Health monitoring systems
- AI-based diagnostic assistants

Future Enhancements

- Mobile application development
- Multilingual support
- Integration with hospital systems (EHR)
- Voice-based explanations
- Personal health tracking and analytics

Results

- Successfully analyzes medical reports automatically
- Accurately identifies normal and abnormal values
- Generates patient-friendly explanations
- Provides condition-based suggestions

References

AI Model's:

- Hugging Face
- Gemini APIs

Output

Medical Report Intelligence and Simplification System

Upload reports, understand findings in simple language, view abnormal values, and compare progress.

1) Upload Report Files

Choose Files 2 files

Upload and Extract

2) Select Report to Analyze

month3_report.pdf (90f51558)

☐ Explain Like I am 10 Years Old

Output Language

English

Analyze Report

Optional: Paste Report Text

Paste report text if you do not want to upload a file...

Analyze Pasted Text

Smart Health Summary

Speak Summary

Smart Health Summary

Speak Summary

Report Type: Diabetes Report

This looks like a diabetes report and your key values are mostly in the normal range.

Report type: Diabetes Report.

Risk level: Low.

Parameters analyzed: 3.

All extracted parameters are within normal range.

Normal parameters: Glucose, Cholesterol, Hemoglobin.

Risk Level: Low

Key Issues

- No major abnormal parameters were detected.

Simple Explanation

Your extracted test values are within the normal range, which is a positive sign.

Why It Matters

Keeping values in the normal range helps reduce long-term risk and supports overall health.

Suggestions

- Continue your current food and activity routine because your values are in the normal range.
- Limit added sugar, deep-fried food, and highly processed snacks to keep glucose and cholesterol stable.
- Do at least 30 minutes of walking or light exercise most days.
- Repeat key tests on schedule to make sure values remain normal.
- Keep regular follow-up with your doctor for preventive monitoring.

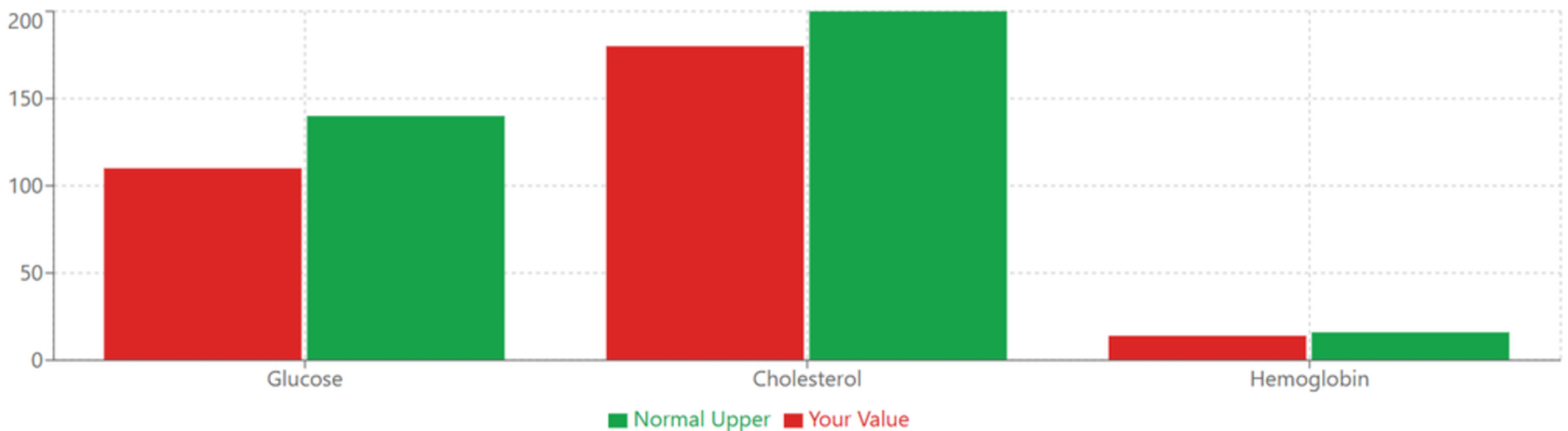
Output

Alerts

Normal vs Your Values

Test	Your Value	Normal Range	Status	Explanation
Glucose	110 m	70.0-140.0 mg/dL	Normal	Your glucose is in the normal range. This is a healthy sign.
Cholesterol	180 m	125.0-200.0 mg/dL	Normal	Your cholesterol is in the normal range. This is a healthy sign.
Hemoglobin	14 g/dL	12.0-16.0 g/dL	Normal	Your hemoglobin is in the normal range. This is a healthy sign.

Bar Chart: Your Values vs Normal Upper Range

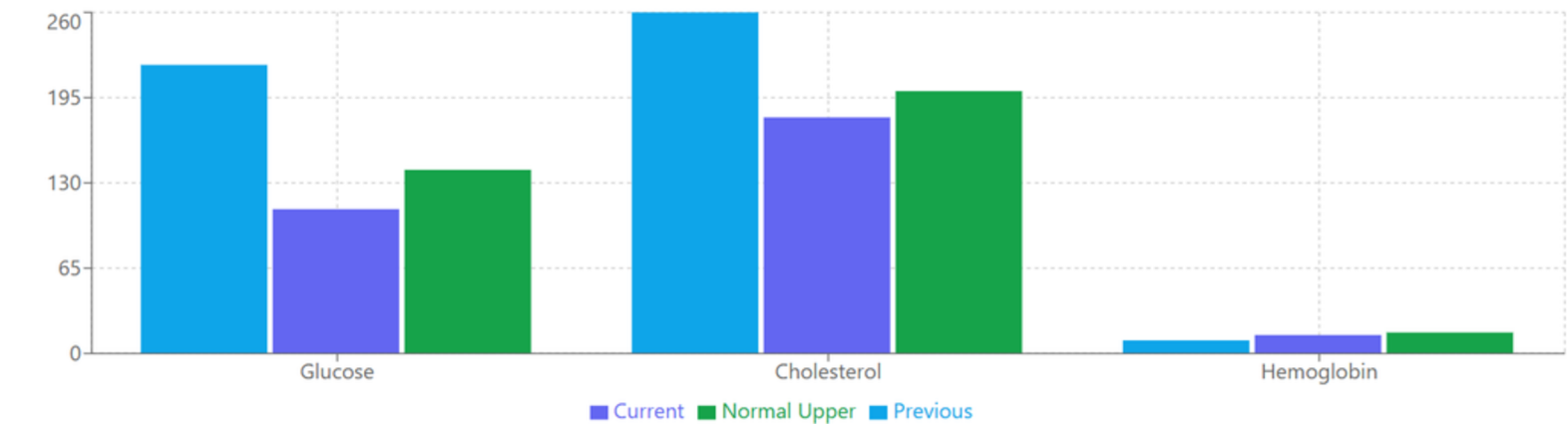


Compare Reports

month1_report.pdf month3_report.pdf Compare Reports

Previous Type: Diabetes Report Current Type: Diabetes Report

Test	Normal Range	Previous	Prev Status	Current	Current Status	Difference	Trend
Glucose	70.0-140.0 mg/dL	220	High	110	Normal	-110	Improved
Cholesterol	125.0-200.0 mg/dL	260	High	180	Normal	-80	Improved
Hemoglobin	12.0-16.0 g/dL	10	Low	14	Normal	4	Improved



Extracted Report Text

Month 3 Report (Recovered / Normal) Patient Details Name: John Doe Age: 45 Condition: Type 2 Diabetes Doctor Summary Patient condition has improved significantly. All major health parameters are now within normal range. Parameter Your Value Normal Range Status Glucose (Fasting) 110 m g/dL 70-140 Normal HbA 1 c 5.5 % 4-5.6 % Normal Cholesterol 180 m g/dL 125-200 Normal Hb 14 g/dL 12-17 Normal

Conclusion

- Simplifies complex medical report understanding
- Bridges gap between medical data and patient knowledge
- Helps users take preventive health actions
- Reduces dependency on expert interpretation

